Can Large Language Models Write Patient Trajectories?

Fine-tuning LLMs on Serialized EHRs for Generating Synthetic Longitudinal EHRs

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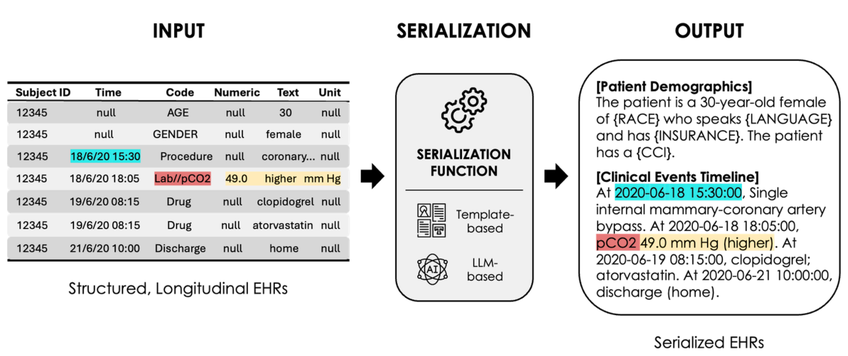
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BACKGROUND

Access to Electronic health records (EHRs) is restricted by privacy concerns, so synthetic EHRs offer a way to share and develop models without exposing real patients.

- **But generating realistic EHRs is hard:** Time is irregular, features are heterogeneous and sparse, and values mix numeric, categorical and binary types. Most existing generators handle only part of this structure.
- **Language models offer a new opportunity:** By writing EHRs as text, one model can learn time, features and values together.

What is ***Serialization***?



Express the full *time*-*feature*-*value* information in natural language

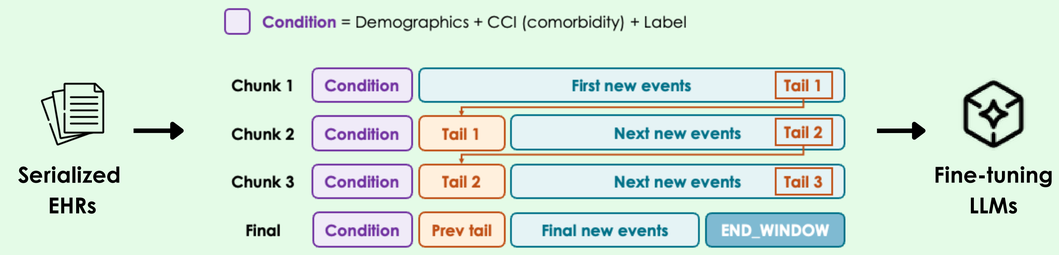
Readable by language models without discarding any information

OBJECTIVES

1. **Generate:** Fine-tune language models on serialized EHRs to generate synthetic longitudinal patient records.
2. **Preserve:** Assess whether synthetic records preserve the realistic values and timing of real patient data.
3. **Usefulness:** Assess whether synthetic records are useful for prediction, by training on synthetic and testing on real data (TSTR) for two prediction tasks.

METHODOLOGY

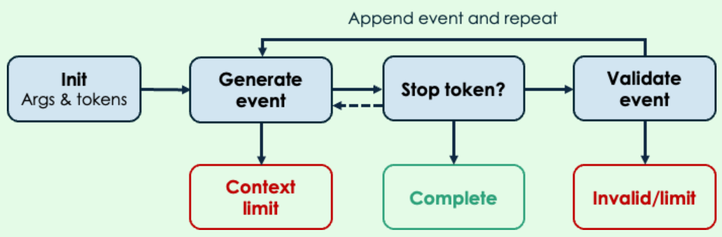
Step 1: Fine-tuning



- Long trajectories were divided into overlapping chunks at complete-event boundaries.
- A pre-trained autoregressive language model was fine-tuned for the prediction task, conditioned on demographics, CCI, and the target outcome label.

Step 2: Synthetic Trajectory Generation

- Events were generated autoregressively using temperature-controlled top-k or top-p sampling.
- A trajectory was considered complete when the model emitted END_WINDOW token.



Experimental Setup

Design choices

- CABG surgery event as the temporal anchor
- Relative-time representation
- Observation window: Events start within [0, 24h)
- Testing 2 models: DistilGPT2 and **BioGPT-Large**

An example of serialized EHR

CONDITION
Age: 70 | Sex: F | CCI: 5 | Readmission_30d: 1

EVENT TIMELINE
At +000900s, [lab] Lactate 3.3 mmol/L (higher).
At +001800s, [drug] Acetaminophen.
At +003600s, [lab] pCO2 45.0 mm Hg (normal).
... ...
<END_WINDOW>

847
Coronary artery bypass grafting (CABG) patients

30-DAY READMISSION
Readmission within 30 days
Positive rate: 11.7%

PROLONGED LENGTH OF STAY
Total hospital stay >7 days (>168 hours)
Positive rate: 19.1%

70 : 10 : 20
TRAIN / VALIDATION / TEST
Patient-level split

Evaluation

Lab value fidelity
- Do synthetic lab values look real?
- KS distance

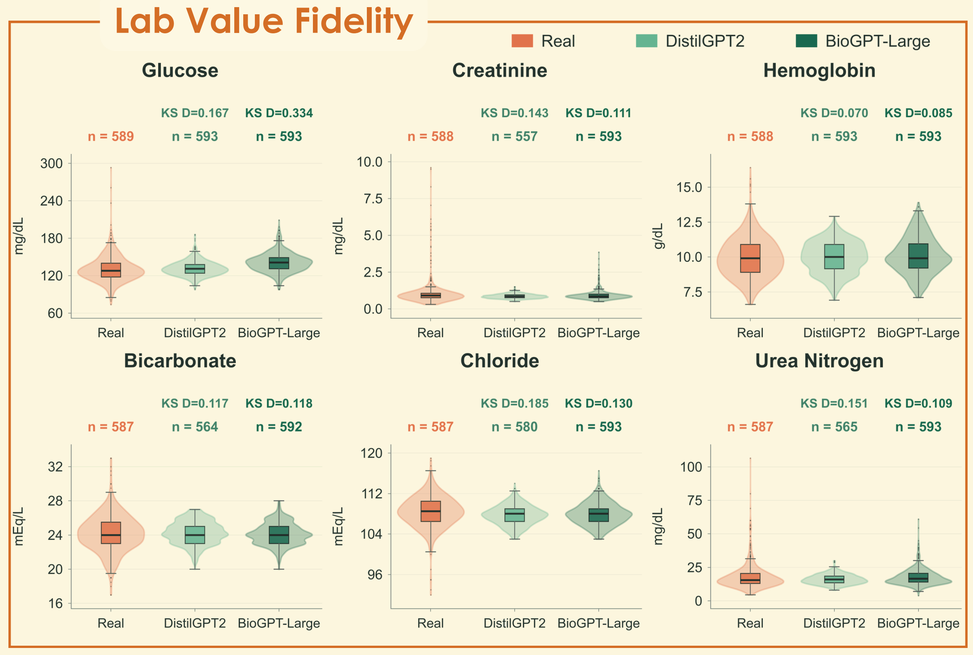
Time fidelity
- Do events happen at realistic times?

Predictive utility
- Is synthetic data useful for prediction?
- TSTR vs. TRTR

RESULTS

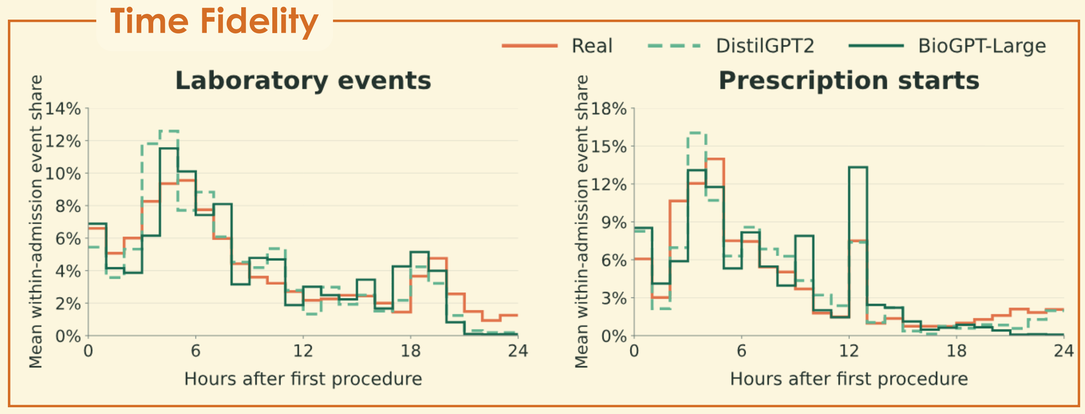
- **Typical values are well captured.** Both models reproduce real medians closely for most labs.
- **Extremes are missing.** Synthetic distributions are narrower than real ones: the long tails that represent the sickest patients are largely absent.

Lab Value Fidelity



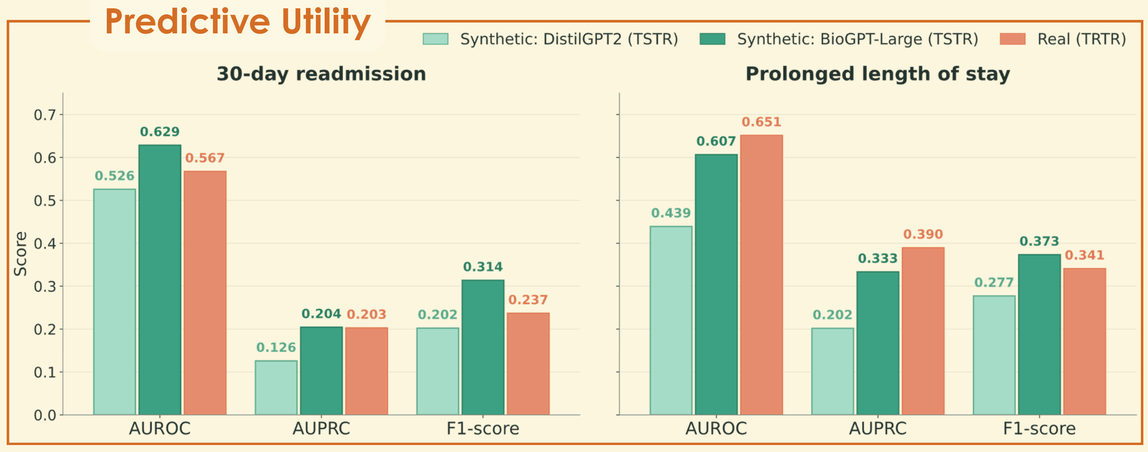
- **The overall temporal pattern is captured.** Both models reproduce the main shape of real data: an early peak, a gradual decline, and a secondary peak.
- **Late-window events are under-generated.** Synthetic labs drop to near zero after ~21h.

Time Fidelity



- **Synthetic data from larger models can outperform real data.** For 30-day readmission, models trained only on BioGPT synthetic records exceed real-data performance (AUROC 0.629 vs. 0.567; F1-score 0.314 vs. 0.237) and match it on AUPRC (0.204 vs. 0.203).

Predictive Utility



CONCLUSIONS

Language models can learn to write patient trajectories. Fine-tuned on serialized EHRs, they generate records that preserve typical lab values, realistic timing, and provide signal for prediction tasks.

They capture the common, not the rare. Across values and time, generators reproduce dominant patterns but under-represent extremes, which are often the most clinically important.

Why can synthetic data match real data? Generators may smooth away rare and noisy cases, producing cleaner patients that are easier to learn from.

- Useful data ≠ Realistic data
- Evaluate clinical plausibility and privacy
- Target the tails



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